



The disease tattoo

Scientists in Switzerland are testing tattoos on mice that only materialise during early stages of a disease. <https://digit.in/tattoomice>



Smartphone microscopic photos

Deep learning can reveal microscopic details in smartphone photos on par with laboratory-grade devices. <https://digit.in/deeplnring>

THE BRAND NEW ORGAN

Have you met the newest organ in the human body?

Arbab Mukherjee | arnab@digit.in

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ith our current state of advancement, you cannot blame humanity at being surprised at the

discovery of a new organ. It's not like we haven't cut open enough human bodies to figure everything out yet, right? Turns out, we have totally overlooked an organ.

Scientists have found a network of fluid filled spaces between tissues that they hadn't seen before. According to a report published in *Scientific Reports*, these fluid filled spaces have been discovered in connective tissue all over the body – under the skin, lining the digestive tract, urinary system and lungs. Meet the human interstitium, the newest organ to be discovered.



A flaw in the normal tissue sample creation process had helped interstitium scale remain elusive all this while.

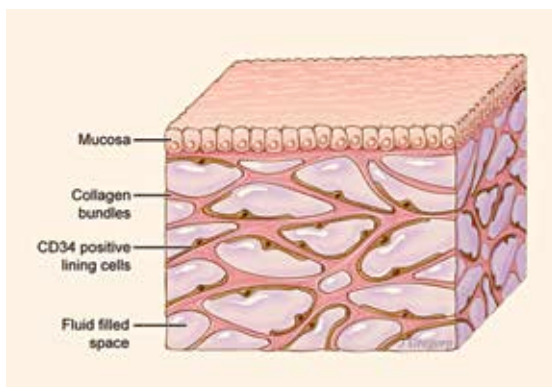
WHAT IS IT?

The human interstitium isn't particularly new. We've been aware of it for ages. To give you an idea, imagine it as what fills the space between cells in the human body – that is what we've known about it so far. Turns out, it is more like a matrix of collagen bundles interspersed with fluid than the densely-packed stacks of connective tissue it appears to be in fixed slides. So, rather than a wall surrounding your cells, it is more like a highway that goes all around them.

Previously, tissue samples had to be explored through fixed slides, which would involve getting a thin cut

of the sample tissue treated chemically first. This would drain out the fluid that made up a major portion of the interstitium. When physicians and coauthors of the paper, Petros Benias and David Carr-Locke used a new technique to monitor and assess patients' bile ducts for cancer spread, they saw something that hadn't been seen before.

The technique, known as "probe-based confocal laser endomicroscopy" or pCLE in short, allowed them to observe living human tissue rather than waiting for the fixed slide. Doing this showed them a net-like pattern in the bile-duct tissue that didn't look like anything anatomical discovery so far. The branches appeared to be 20µm wide. Upon further examination, the coauthors, along with Neil Theise, a pathologist at New York University's



Interstitium in a diagram